

EEE 419/591
Summ 2026
Final Project – Missing-Data Large Datasets

Introduction

In the midterm project, you developed a code to deal with large datasets without loading them into the memory entirely. You worked on each chunk of data (almost) independently. In this project, the dataset you are given has several missing data. The data is missing completely at random (MCAR), which is relatively easier to deal with even if it is not as common in real-world datasets. The question you try to answer in this project is “What is the best algorithmic methodology to train a classifier with a large dataset but with missing data?”. The reason this is a challenging question is because using feature-selection methods you developed in the midterm do not apply directly to datasets with missing data. Alternatively, using off-the-shelf algorithms that work with missing data is not applicable directly as it depends on loading the entire dataset into the memory, something that is not feasible with large datasets.

In this project, you will write a single script with two methods and compare their performance. The two methods are “Imputation-Based” and “Imputation-Free”. Hopefully, by the end of this project, you will understand which one is more suitable for this MCAR type of dataset.

Problem Statement

In this project, you are asked to write a complete Python code that classifies the multi-class vertically partitioned dataset provided. The dataset you are providing with has some missing data. Your single script needs to evaluate two methods: 1) Imputation-Based; and 2) Imputation-Free. The two are explained below. Your code will need to not only compare the two numerically to each other but will also print to the grader 3 comment paragraphs that you need to write: “Imputation-Based Justification”, “Imputation-Free Justification”, and “Conclusion”. These paragraphs explain your understanding (more on this below). You are allowed to use any idea(s) you deem fit to achieve this goal as long as it does not involve loading the entire dataset into the RAM all at once.

1) Imputation-Based Method:

Replacing the missing data in the given dataset with actual data instead of nan’s is helpful. It allows using conventional ML algorithms that work on complete datasets with no missing data. However, if the imputation method is not carefully chosen, you would be confusing the trainer more than you are helping it, deteriorating accuracy. You are allowed to use any imputation method you deem fit as long as you justify it in the “Imputation-Based Justification”. A single paragraph with only a few sentences is needed in “Imputation-Based Justification”. Answers to some of the following questions are examples of what you might write:

- a. What imputation-based algorithms have you tried? What was the accuracy of each? Either try only 1 method while providing its citation in a published paper/book (advanced) or try at least 2 well-thought imputation-based algorithms and compare them to each other. Your submitted code must implement only 1 imputation-based algorithm; the one you recommend.
- b. What made you pick the imputation-based algorithm you submitted? Published somewhere? High accuracy? Low computational time? A mixture of all?
- c. What parameters, if any, have you tuned for the method you submitted?
- d. What is the resulting change in accuracy when you made the parameter tuning? Increase? Decrease? Was this expected? If no, explain where the inconsistency might be coming from.

Since you are not allowed to load the entire dataset in the RAM, you might use the algorithm you submitted for the midterm or any other algorithm you deem fit.

2) **Imputation-Free Method:**

Instead of imputing the dataset, suggest a method to work on the dataset as is. Write the code for that method. Since the dataset is large with missing data, neither using the algorithms you used in the midterm (PCA, feature selection... etc.) nor implementing an imputation-free algorithm that is in the literature “as-is” would not work. You will need to either modify it or come up with a new algorithm. In the “Imputation-Free Justification” paragraph, provide a justification to your choice. Answers to some of the following questions are examples of what you might write:

- a. What is the high-level idea that your algorithm is based on?
- b. How does your algorithm circumvent the missing data problem?
- c. Have you tried other algorithms? If yes, what made you pick this algorithm. If no, explain why? Execution time? Performance? Else?
- d. What parameters, if any, have you tuned for the method you submitted?
- e. What is the resulting change in accuracy when you made the parameter tuning? Increase? Decrease? Was this expected? If no, explain where the inconsistency might be coming from.

3) **Conclusion:**

Compare the performance of both methods. Print the accuracy and runtime of each algorithm then write a “Conclusion” comment. Answers to some of the following questions are examples of what you might write:

- a. Which of the two methods outperformed the other?
- b. Is this expected? If yes, justify? If no, why? And where do you suspect the problem to be?

Dataset Properties

Core Properties

The dataset provided has the following parameters:

- Number of Features: 10,000
- Number of training dataset points: 70,000
- Number of testing dataset points: 30,000
- Number of classes: 100 (labelled 0-99)
- **Percentage of dataset points with missing data: 99.99%**
- **Percentage of features with missing data in each row: 50%**
- **Type of data missing data: Missing completely at random (MCAR)**
- **Values of missing data: *np.nan***
- **Dataset has missing labels: No**

Dataset Structure

Each of the training and testing portions is further split into 10 chunks, each chunk spans 1,000 consecutive features. Each chunk is stored in a separate file named according to its index bounds and whose name ends in “*.parquet*” extension. Further, the label (i.e. class) column for each is provided in a separate file named “*labels_param.parquet*”.

Dataset Acquiring [No changes from the midterm, except for the download link and the alternative dataset generation code]

You have two options to acquire the dataset for this project:

1. Download the pre-generated dataset from the following link (~3.3GB):
https://www.dropbox.com/scl/fo/dzawpsg20002kqajxqw0m/AE07DY_t4900UR2DteT0Y98?rlkey=647glfmxhipdtkssakkinp25r&st=gkjinqhi2&dl=0
2. Run the provided dataset generation script *final_dataset_gen.py* on your machine without changing any parameters (10-15 mins to run). Do not modify the generation code or parameters at all. Otherwise, your script will run on a different dataset than what the grader is expecting, resulting in a failed evaluation.

After the dataset is saved on your computer, you can load any of its files in your code either via the example commands provided to you in the project *final_project.py* template, or any other commands you deem fit. For the provided commands to work, make sure that your dataset directory structure looks like this:

Parent_directory < where your to-be-developed project file is located

```
| final_dataset/  
| | — train/  
| | | — labels_param.parquet <-- Contains 'class_label' & encoded metadata  
| | | — features_0_1000.parquet <-- Feature Chunk 1 (Columns 0-999)
```

```

| | | — features_1000_2000.parquet <-- Feature Chunk 2 (Columns 1000-1999)
| | | — ... <-- Remaining Feature Chunks
| | | — test/
| | | — labels_param.parquet <-- Contains 'class_label' & encoded metadata
| | | — features_0_1000.parquet <-- Feature Chunk 1 (Columns 0-999)
| | | — features_1000_2000.parquet <-- Feature Chunk 2 (Columns 1000-1999)
| | | — ... <-- Remaining Feature Chunks

```

Technical Requirements

Your code must meet the following requirements: [No changes from the midterm]

1. **RAM Constraint:** Your solution **MUST NOT** load the entire dataset into the RAM all at once. At most, your code may only hold 25% of the total dataset features in active memory at any given moment. Students who submit a code that loads the entire dataset in the RAM will automatically receive a grade of zero.
2. **Classification:** Your code must be an ML-based algorithm. Non-ML-based codes will not be accepted.
3. **Runtime Limit:** Your entire script must be optimized. Codes that take more than 40 minutes to run on your own computer will not be accepted. It will typically take far less than that if run on an average computer in the market.
4. **Performance Expectations:** The classification performance does not need to be perfectly optimal. You will not be graded on the absolute accuracy score unless your result is too far (>10% lower) from the maximum accuracy reported on Ed Discussion by your peers.

Deliverables [Changes from the midterm are in bold font]

- **Use the provided template script.**
- You must submit a single Python file named ***Final_Project_FirstName_LastName.py***.
- Do not submit the dataset.
- Your code must expect the dataset to be in the same format and name it is provided
- Your code must print out the final validation accuracy and the total execution time **for each method as well as the 3 comment paragraphs** directly to the terminal.
- No written reports or documentation are needed. If you experimented with multiple machine learning algorithms or parameters that did not work as expected, do not include them in your final code submission. Only the single algorithm that successfully works **for each method that prints the accuracy, execution time and the 3 comment paragraphs** are the deliverables.
- **Make sure the 3 comment paragraphs are printing nicely to the terminal. You will manually “hard code” each of them as a string in your script, but you can use AI to help add break lines to read nicely and not be printed on a single line.**

Policies

[Changes from the midterm are in bold font]

Allowed:

1. The use of AI (ChatGPT...etc.), especially if you use it to:
 - Ask it to recommend specific commands that perform a task.
 - Paste parts of your non-working code along with error messages to understand why it is failing.
 - *Note: You should not use AI suggestions blindly since AI can hallucinate.*
2. Asking AI anything.
3. The use of any external resources (internet, textbooks, instructional videos, forums).
4. Discussions through the Ed Discussion board with any question/answer/code (except sharing your entire code). This is highly encouraged to deepen your understanding. Reminder that it is best practice to limit your screenshots or code shares to the commands that deliver the message you want to deliver as opposed to sharing a lot and expecting the reader to read everything and figure it out themselves.
5. Sharing final answers (e.g., your final accuracy percentage or total execution runtime). Make sure you don't create a new post for this if one has already been created.
6. Sharing ranges of parameter values you tried or recommend.
7. Sharing the name of the classification or preprocessing algorithms you plan to submit, work on, or have (not) tried and, more encouragingly, why you (not) recommend them.
8. Sharing screenshots of codes that don't work or give errors to ask your peers for feedback.
9. Sharing fundamental conceptual ideas that would work or would not work.
- 10. Using the code you submitted in the midterm.**

Not Allowed:

1. Submitting the entire code that an AI gives you without understanding what it does. AI is allowed because it helps you understand and be productive, but keep in mind that, by the end of the day, it is still a "robot". You are responsible for what you submit.
2. Seeking help from any human being outside this class (private tutors, friends, family, etc.).
3. Meeting to work on the entire project (or even a major portion of it) together. This project is an individual effort. Discussion and asking specific questions over Ed Discussion, regardless of how much information they reveal, however, is completely allowed and even encouraged.
- 4. Sharing with your peers the verbatim of any of the 3 comment paragraphs you plan to submit. Sharing the content of any of these paragraphs is, however, allowed and even encouraged.**

MANDATORY – Reminder to cite your AI chat. Check the syllabus document for more details